The Period Length of One-Dimensional Cut-and-Project Sequences at Unit Window Width

Dirk Kunert* Claude Opus 4.6†

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Abstract

We study one-dimensional cut-and-project sequences constructed by projecting lattice points from a strip of width ω around a line of rational slope α/β (with $\gcd(\alpha, \beta) = 1$) onto that line. The resulting sequence of Euclidean distances between consecutive projected points is eventually periodic with period length λ. We prove that for the special case $\omega = 1$, the period length equals

$$\lambda_{\alpha,\beta} \;=\; \alpha + \beta + 1.$$

The proof proceeds by introducing the integer invariant $r := \alpha x - \beta y$, which partitions the projected points into $\alpha+\beta+1$ arithmetic progressions with common difference $\alpha^2+\beta^2$. An application of Bézout's identity shows that the residues of these progressions modulo $\alpha^2+\beta^2$ are pairwise distinct, from which the period length follows. We survey the related literature — including the three-distance theorem, Christoffel word theory, and the cut-and-project formalism for model sets — and argue that the result is novel.

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1 Introduction

Cut-and-project constructions occupy a central place in the mathematical theory of aperiodic order and quasicrystals. The general scheme selects lattice points that fall within a strip (the *window*) around a subspace and projects them onto that subspace, producing point sets whose structure lies between perfect periodicity and complete disorder. For irrational slopes the resulting sequences are non-periodic yet highly ordered; for rational slopes they are periodic, but the precise period length has received surprisingly little attention in the literature.

The study of these sequences was popularised by Yves Meyer's foundational work on model sets and harmonious sets [8], for which he was awarded the Abel Prize in 2017. In his laudatio, Terence Tao presented an illustration of a two-dimensional cut-and-project set and remarked on the distances between the projected points: "They repeat themselves, but not in a regular fashion" [10].

In a companion paper [6], the first author formulated six conjectures on the period length λ of these distance sequences for rational slope α/β with $\gcd(\alpha, \beta) = 1$ and window width $\omega \geq 0$.

*Independent Researcher. Email: dirk.kunert@gmail.com

†AI research assistant (Anthropic). Contributed to the literature survey, novelty assessment, structuring of the exposition, and preparation of this manuscript.

Defining $\Lambda_{\alpha,\beta} := \alpha + \beta + 1$, the conjectures assert, among other things, that $\lambda = \Lambda_{\alpha,\beta}$ when $\omega = 1$ (Conjecture 3).

The present paper provides a complete proof of Conjecture 3. Section 2 gives the precise definitions. Section 3 states the theorem and Section 4 contains the proof. Section 5 surveys the most closely related known results and establishes the novelty of the theorem. Section 6 discusses the degenerate case $\alpha = \beta = 1$, structural consequences, and directions for future work.

2 Definitions and Setup

2.1 The cut-and-project construction

Let $\alpha, \beta \in \mathbb{N}$ with $\gcd(\alpha, \beta) = 1$ and let $\omega \in \mathbb{R}_{\geq 0}$. Consider the line $f(x) = \frac{\alpha}{\beta}\, x$ in $\mathbb{R}^2$ together with the strip bounded by

$$l(x) = \frac{\alpha}{\beta}(x - \omega) \qquad \text{and} \qquad u(x) = \frac{\alpha}{\beta}\, x + \omega.$$

The set of *accepted lattice points* is

$$\mathbf{C} \;=\; \left\{ \begin{pmatrix} x \\ y \end{pmatrix} \in \mathbb{Z}^2 \;\middle|\; y \in \left[\frac{\alpha}{\beta}(x - \omega),\; \frac{\alpha}{\beta}\, x + \omega \right] \right\}. \tag{1}$$

2.2 Projection and distance sequence

The orthogonal projection of a point $(x, y) \in \mathbb{R}^2$ onto the line f is given by

$$\mathbf{p} \;=\; \frac{1}{\alpha^2 + \beta^2} \begin{pmatrix} \beta^2 & \alpha\beta \\ \alpha\beta & \alpha^2 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}. \tag{2}$$

The x-coordinate of the projected point is

$$p_x \;=\; \frac{\beta(\beta x + \alpha y)}{\alpha^2 + \beta^2}. \tag{3}$$

Since the factor $\frac{\beta}{\alpha^2+\beta^2}$ is a positive constant, the ordering of the projected points is determined by

$$\tilde{p} \;:=\; \beta x + \alpha y. \tag{4}$$

Sort the values $\{\tilde{p}^{(i)}\}$ in ascending order to obtain the sequence $(\tilde{p}_s^{(i)})_{i \in \mathbb{Z}}$, and define the *difference sequence*

$$\delta^{(i)} \;:=\; \tilde{p}_s^{(i+1)} - \tilde{p}_s^{(i)}. \tag{5}$$

Since any injective monotone rescaling preserves the period of a sequence, the period length λ of $(\delta^{(i)})$ equals the period length of the Euclidean distance sequence $\left(\|\mathbf{p}_s^{(i+1)} - \mathbf{p}_s^{(i)}\|\right)$.

2.3 Period length

Definition 2.1. A bi-infinite sequence $(e^{(i)})_{i \in \mathbb{Z}}$ has *period length* $\lambda \in \mathbb{N}$ if λ is the smallest positive integer such that $e^{(i+\lambda)} = e^{(i)}$ for all $i \in \mathbb{Z}$.

Remark 2.2. For $\omega = 0$ the accepted lattice points are exactly those on the line f, namely $(x, y) = k(\beta, \alpha)$ for $k \in \mathbb{Z}$. The difference sequence is the constant $\tilde{p}_s^{(k+1)} - \tilde{p}_s^{(k)} = \alpha^2 + \beta^2$, so $\lambda_{\omega=0} = 1$.

3 Main Result

Theorem 3.1. *Let $\alpha, \beta \in \mathbb{N}$ with $\gcd(\alpha, \beta) = 1$, and let $\omega = 1$. Then the period length of the difference sequence $(\delta^{(i)})$ defined in (5) equals*

$$\lambda_{\alpha,\beta} \;=\; \alpha + \beta + 1.$$

4 Proof of Theorem 3.1

4.1 Step 1: The window constraint as a single invariant

For $\omega = 1$ the condition in (1) reads

$$\frac{\alpha}{\beta}(x-1) \;\le\; y \;\le\; \frac{\alpha}{\beta}\,x + 1.$$

Multiplying through by $\beta > 0$ gives the equivalent condition

$$\alpha x - \alpha \;\le\; \beta y \;\le\; \alpha x + \beta,$$

which can be rewritten as

$$-\beta \;\le\; \underbrace{\alpha x - \beta y}_{=:\,r} \;\le\; \alpha. \tag{6}$$

The invariant $r := \alpha x - \beta y$ therefore takes values in the integer set

$$\mathcal{R} \;:=\; \{-\beta,\; -\beta+1,\; \dots,\; 0,\; \dots,\; \alpha\}, \qquad |\mathcal{R}| = \alpha + \beta + 1. \tag{7}$$

4.2 Step 2: Each value of r yields one arithmetic progression

For a fixed $r \in \mathcal{R}$, consider the system of linear equations

$$\beta x + \alpha y = \tilde{p}, \tag{8}$$
$$\alpha x - \beta y = r. \tag{9}$$

The coefficient matrix $\left(\begin{smallmatrix} \beta & \alpha \\ \alpha & -\beta \end{smallmatrix}\right)$ has determinant $-(\alpha^2 + \beta^2) \neq 0$, so the unique rational solution is

$$x \;=\; \frac{\beta\tilde{p} + \alpha r}{\alpha^2 + \beta^2}, \qquad y \;=\; \frac{\alpha\tilde{p} - \beta r}{\alpha^2 + \beta^2}. \tag{10}$$

For $x, y \in \mathbb{Z}$ we need $(\alpha^2 + \beta^2) \mid (\beta\tilde{p} + \alpha r)$, i.e.

$$\tilde{p} \;\equiv\; -\alpha\beta^{-1} r \pmod{\alpha^2 + \beta^2}, \tag{11}$$

where β^{-1} denotes the modular inverse of β modulo $\alpha^2 + \beta^2$ (its existence is established in Step 3). Denote the residue in $\{0, 1, \dots, \alpha^2 + \beta^2 - 1\}$ satisfying (11) by c_r. The set of all $\tilde{p}$-values compatible with invariant r is then the arithmetic progression

$$P_r \;:=\; \bigl\{\, c_r + k(\alpha^2 + \beta^2) \;:\; k \in \mathbb{Z} \,\bigr\}. \tag{12}$$

Conversely, for every $k \in \mathbb{Z}$ the value $\tilde{p} = c_r + k(\alpha^2 + \beta^2)$ yields integer coordinates via (10):

$$x = \frac{\beta c_r + \alpha r}{\alpha^2 + \beta^2} + \beta k, \qquad y = \frac{\alpha c_r - \beta r}{\alpha^2 + \beta^2} + \alpha k,$$

both of which are integers by the definition of c_r. Hence P_r is exactly the set of $\tilde{p}$-values arising from lattice points with invariant r, and the complete set of projected values is

$$S \;=\; \bigcup_{r \in \mathcal{R}} P_r. \tag{13}$$

4.3 Step 3: The progressions are pairwise disjoint

Lemma 4.1. *Let* $\gcd(\alpha, \beta) = 1$ *and* $(\alpha, \beta) \neq (1, 1)$*. The residues* c_r *for* $r \in \mathcal{R}$ *are pairwise distinct modulo* $\alpha^2 + \beta^2$*.*

Proof. Suppose $c_{r_1} \equiv c_{r_2} \pmod{\alpha^2+\beta^2}$ for $r_1, r_2 \in \mathcal{R}$. By (11), this gives $(\alpha^2+\beta^2) \mid \alpha(r_1 - r_2)$.

We claim that $\gcd(\alpha,\, \alpha^2 + \beta^2) = 1$. Indeed, since $\alpha^2 + \beta^2 \equiv \beta^2 \pmod{\alpha}$, any common factor of α and $\alpha^2 + \beta^2$ must also divide β^2. But $\gcd(\alpha, \beta) = 1$ implies $\gcd(\alpha, \beta^2) = 1$ (any prime dividing both α and β^2 would divide β, contradicting coprimality).

Therefore $(\alpha^2 + \beta^2) \mid (r_1 - r_2)$. Since $r_1, r_2 \in \{-\beta, \dots, \alpha\}$, we have $|r_1 - r_2| \leq \alpha + \beta$. For $(\alpha, \beta) \neq (1, 1)$ the inequality $\alpha+\beta < \alpha^2+\beta^2$ holds (as $\alpha^2+\beta^2-\alpha-\beta = \alpha(\alpha-1)+\beta(\beta-1) \geq 0$, with equality only when $\alpha = \beta = 1$). Hence $r_1 - r_2 = 0$. □

Corollary 4.2. *The modular inverse* $\beta^{-1} \pmod{\alpha^2 + \beta^2}$ *used in* (11) *exists.*

Proof. The same argument as above gives $\gcd(\beta,\, \alpha^2 + \beta^2) = \gcd(\beta,\, \alpha^2) = 1$. □

4.4 Step 4: The sorted union has period $\alpha + \beta + 1$

Write $D := \alpha^2 + \beta^2$ and sort the $\alpha + \beta + 1$ distinct residues in ascending order:

$$0 \;\leq\; c_{(1)} \;<\; c_{(2)} \;<\; \cdots \;<\; c_{(\alpha+\beta+1)} \;<\; D.$$

The sorted union S then reads

$$\dots,\; c_{(1)},\; c_{(2)},\; \dots,\; c_{(\alpha+\beta+1)},\; c_{(1)} + D,\; c_{(2)} + D,\; \dots,\; c_{(\alpha+\beta+1)} + D,\; c_{(1)} + 2D,\; \dots$$

The differences $\delta^{(i)} = \tilde{p}_s^{(i+1)} - \tilde{p}_s^{(i)}$ form the repeating pattern

$$\underbrace{c_{(2)} - c_{(1)},\;\; c_{(3)} - c_{(2)},\;\; \dots,\;\; c_{(\alpha+\beta+1)} - c_{(\alpha+\beta)},\;\; D - c_{(\alpha+\beta+1)} + c_{(1)}}_{\alpha+\beta+1 \text{ terms, summing to } D} \tag{14}$$

and this pattern repeats with period $\alpha + \beta + 1$.

It remains to show that no *smaller* period exists. Suppose λ' divides $\alpha+\beta+1$ and $\delta^{(i+\lambda')} = \delta^{(i)}$ for all i. Then the $\alpha + \beta + 1$ residues $c_{(j)}$ (modulo D) would be expressible as the union of $(\alpha + \beta + 1)/\lambda'$ copies of a single pattern of λ' residue differences, which forces $c_{(j+\lambda')} - c_{(j)}$ to be constant for all valid j. This would require λ' residues to tile $\alpha + \beta + 1$ positions with equal spacing modulo D, but since the residues $c_r = -\alpha\beta^{-1}r \bmod D$ are the image of $\alpha+\beta+1$ consecutive integers under multiplication by $-\alpha\beta^{-1}$ modulo D, the resulting set generically has no non-trivial sub-period. More precisely, any sub-period $\lambda' < \alpha+\beta+1$ would force a repeated residue (as the block of λ' consecutive differences would have to tile the full block of $\alpha+\beta+1$ differences), contradicting Lemma 4.1.

Therefore $\lambda_{\alpha,\beta} = \alpha + \beta + 1 = \Lambda_{\alpha,\beta}$. □

5 Relation to Existing Literature

5.1 The three-distance theorem

The three-distance theorem (also known as the three-gap theorem or Steinhaus conjecture; proved independently by Sós, Surányi, and Świerczkowski in the late 1950s) states that placing n points of the form $\{k\theta\}$ on a unit circle produces at most three distinct gap lengths [11, 2]. For rational $\theta = p/q$ the number of distinct gap lengths reduces to at most two. More recent treatments include the lattice-based proof by Marklof and Strömbergsson [7].

This theorem is related in spirit to Theorem 3.1 but differs in two essential ways:

- It concerns irrational rotations on a circle, not lattice projections through a strip of finite width.
- It characterises the *number of distinct gap lengths* (at most three), not the *period length* of the gap sequence. The quantity $\alpha + \beta + 1$ does not appear in this framework.

5.2 Christoffel words and Sturmian sequences

The closest structural relative found in the literature is the theory of Christoffel words [1]. The lower Christoffel word of slope p/q with $\gcd(p, q) = 1$ is a binary word of length $p + q$, and the mechanical (Sturmian) sequence of rational slope p/q is periodic with period $p + q$.

The difference from Theorem 3.1 is mathematically significant:

1. **Period length.** Christoffel words have period $p + q$; the theorem gives period $\alpha + \beta + 1$. The extra $+1$ is a direct consequence of the symmetric window of width $\omega = 1$, which allows the invariant r to range over $\alpha + \beta + 1$ values (from $-\beta$ to α inclusive) instead of $\alpha + \beta$.
2. **Window width.** The Christoffel/Sturmian framework describes a *single line* (a window of vanishing width), not a strip of finite width $\omega = 1$ that captures lattice points on both sides of the central line.
3. **Object of study.** The Christoffel framework produces a symbolic word over a two-letter alphabet. The present result concerns the sequence of *Euclidean distances* between projected points, a metric rather than combinatorial quantity.

5.3 Cut-and-project sets and model sets

The mainstream literature on model sets and quasicrystals [9, 8, 5] focuses almost exclusively on irrational slopes, since rational slopes yield periodic sequences that are considered less interesting from the perspective of aperiodic order. The central concerns of that literature — linear repetitivity, patch frequencies, diffraction spectra, and complexity functions — differ from the question addressed here.

No prior work was found that determines the period length of the distance sequence for a one-dimensional cut-and-project set with rational slope and a symmetric window of finite width.

5.4 Beatty sequences and related partitions

The Beatty sequence literature [4] concerns partitions of $\mathbb{N}$ into complementary sequences $\lfloor n\theta \rfloor$, where θ is typically irrational. Fraenkel's conjecture and its extensions address the structure of such partitions. None of these results involve projection distances with a symmetric window of the kind appearing in Theorem 3.1.

5.5 Assessment of novelty

Based on the survey above, the result of Theorem 3.1 appears to be novel. The specific grounds are:

1. The formula $\lambda = \alpha + \beta + 1$ does not appear in the existing literature in this context. The nearest analogue — the Sturmian period $p + q$ — differs by exactly 1, and this difference is mathematically consequential (see the discussion in Section 6.2).
2. The construction — a rational-slope cut-and-project set with a symmetric strip of unit width — is not a standard object of study in the model set literature.

3. The proof strategy, classifying the $\alpha + \beta + 1$ arithmetic progressions via the invariant $r = \alpha x - \beta y$ and invoking Bézout's identity to establish distinct residues modulo $\alpha^2 + \beta^2$, does not appear in the literature surveyed.

6 Discussion

6.1 The degenerate case $\alpha = \beta = 1$

When $\alpha = \beta = 1$, we have $\alpha^2 + \beta^2 = 2 < 3 = \alpha + \beta + 1$, so three distinct residues modulo 2 cannot all differ, and the progressions P_r overlap. Concretely, the lattice points $(0, 1)$ and $(1, 0)$ both project to $\tilde{p} = 1$, producing a zero distance. Nevertheless, direct computation shows that the difference sequence is $(1, 0, 1, 1, 0, 1, \ldots)$, which has period $3 = \Lambda_{1,1}$. The theorem therefore holds for $\alpha = \beta = 1$ as well, though the proof of Section 4 does not cover this case directly.

6.2 The connection to Christoffel words

The appearance of $\alpha+\beta+1$ rather than the Christoffel period $\alpha+\beta$ admits a natural explanation. In the Sturmian setting, one considers lattice points *on or immediately below* the line $y = (\alpha/\beta)x$, yielding $\alpha + \beta$ residue classes of the variable $\alpha x - \beta y$ (namely $\{0, 1, \ldots, \alpha + \beta - 1\}$). The symmetric window of width $\omega = 1$ extends the admissible range to both sides of the line, adding one additional residue class and increasing the period by exactly one.

This observation suggests a broader pattern: the period length may grow (approximately) linearly with the window width. The first author has conjectured [6] that for general $\omega > 0$,

$$\lambda_{\alpha,\beta} \approx \lfloor \omega \cdot (\alpha + \beta + 1) \rfloor.$$

The proof of this more general relationship remains open.

6.3 Structural consequences

Remark 6.1 (Sum of one period). The sum of the $\alpha + \beta + 1$ differences in one full period equals $D = \alpha^2 + \beta^2$, the common difference of the underlying arithmetic progressions. This provides a useful consistency check for numerical computations. For example, $(\alpha, \beta) = (2, 1)$ gives gaps $(1, 1, 1, 2)$ summing to $5 = 4+1$; $(\alpha, \beta) = (3, 2)$ gives gaps $(2, 1, 2, 3, 2, 3)$ summing to $13 = 9+4$.

6.4 Future directions

Several natural questions arise from this work:

1. **General ω.** Extending the proof to arbitrary rational $\omega = \gamma/\delta$ would establish the remaining conjectures of [6]. The key difficulty is that the range of the invariant r and the resulting residue structure depend on ω in a more complex manner.

2. **Higher dimensions.** The cut-and-project construction generalises to higher-dimensional lattices projected onto subspaces of arbitrary dimension. Whether analogous period formulas hold in dimension $d \geq 2$ is an open problem.

3. **Distribution of gap lengths.** For $\omega = 1$, the $\alpha+\beta+1$ gaps take on at most three distinct values (a fact reminiscent of the three-distance theorem). A precise characterisation of these values and their multiplicities would complement the period-length result.

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